

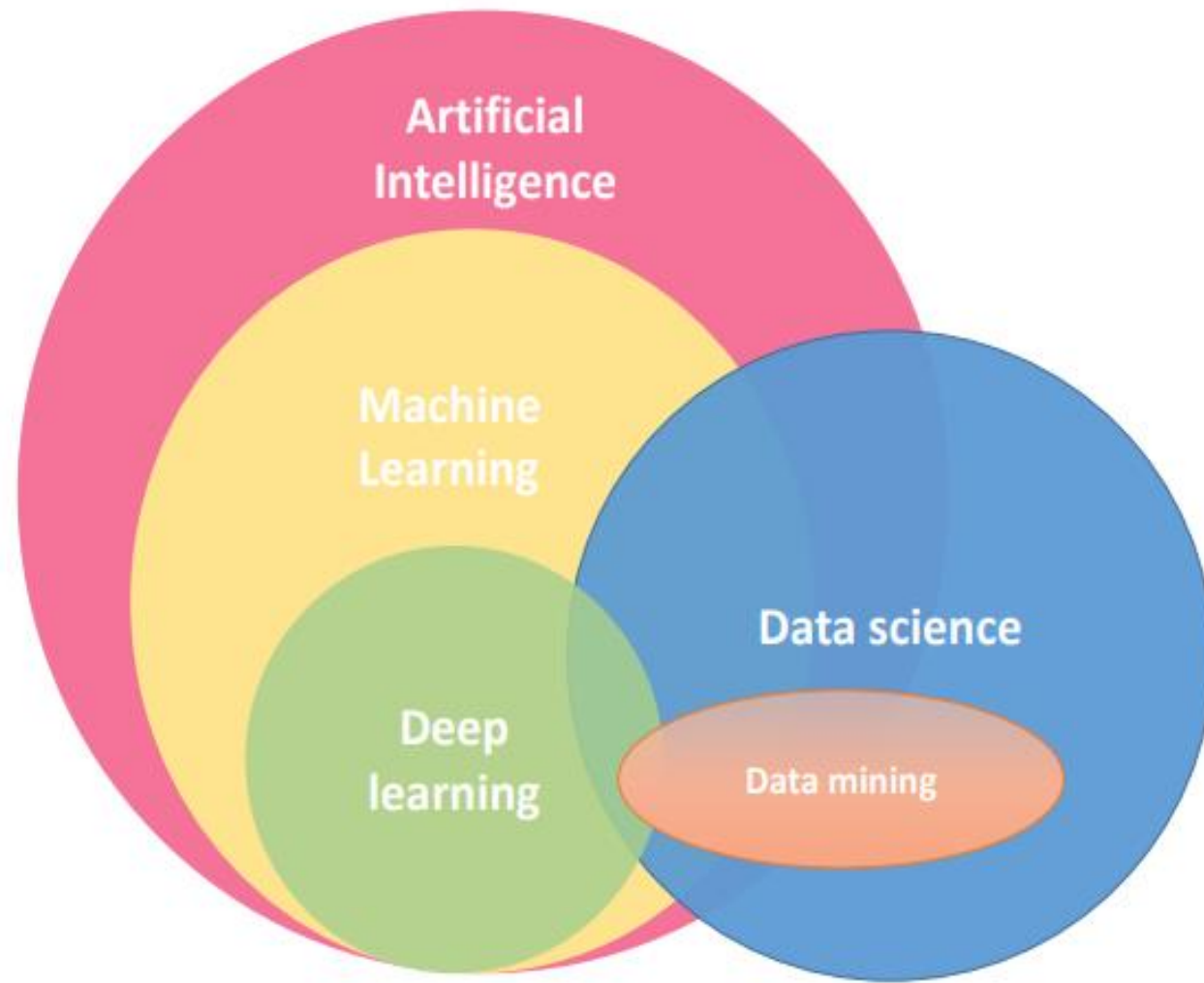
Data Mining

INTRODUCTION

LAB 1

Types of Data Analysis





Artificial Intelligence

Artificial intelligence (AI) is concerned with **making machines smart** aiming to create a **system which behaves like a human**.

AI solve problem and perform tasks that needs Human Intelligence in order to make decisions.

This involves fields such as robotics, natural language processing, information retrieval, computer vision and machine learning.

Machine learning

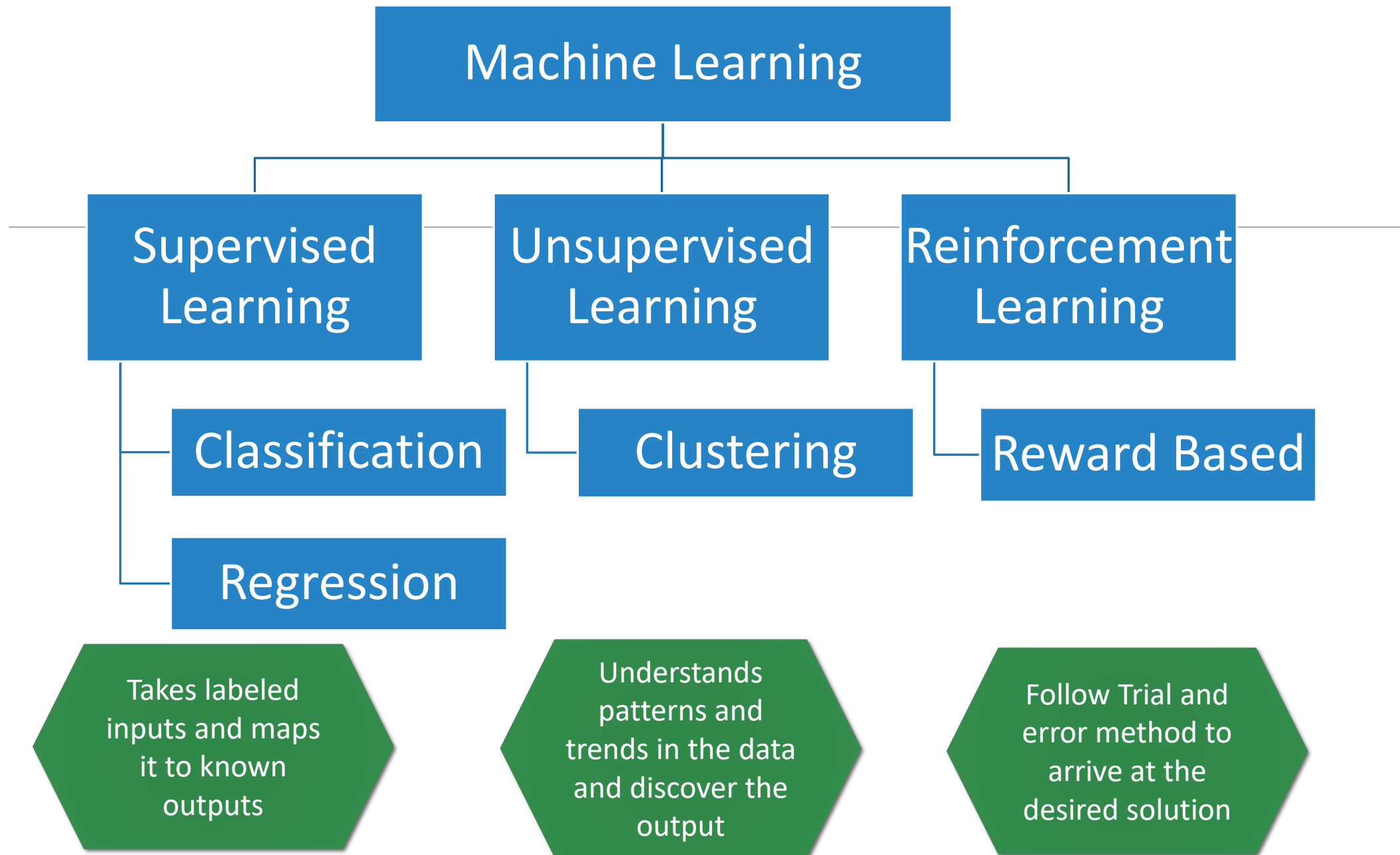
Machine learning (ML) is a **subset of AI**.

ML imitates the way of human thinking and gradually improve the accuracy.

ML aims to **develop algorithms that can learn** from historical data and **improve the system with experience**.

Human provide features to learn the machine.

ML is the step about taking the cleaned/transformed data and **predicting future outcomes**.



Deep learning

Deep learning is a subset of ML, Inspired by Human brain

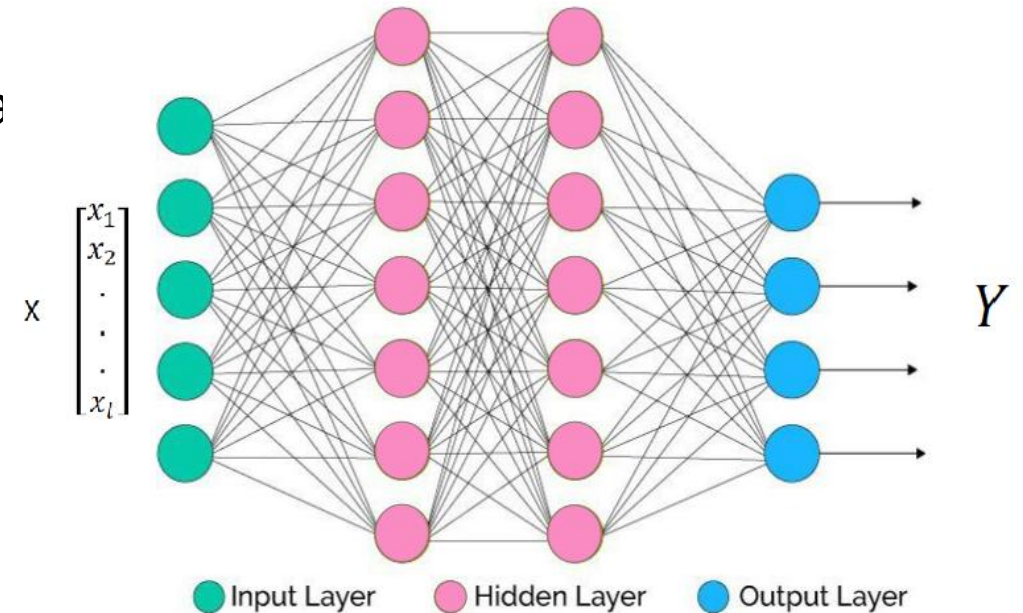
Data is passed via *multiple number of non-linear* transformations to calculate an output.

The term *deep* refers to many steps in this ca

Observe and learn by itself using ANN

Deep learning is

- A machine learning at large scale.
- With more sophisticated neural networks.
- Works with larger dataset.



Big Data

Big Data refers to a **huge volume of data** of various types, i.e., structured, semi structured, and unstructured.

This data is generated through various digital channels such as mobile, Internet, social media, e-commerce websites, etc

Velocity refers to the speed of processing the data and analyze it.

Volume massive of data

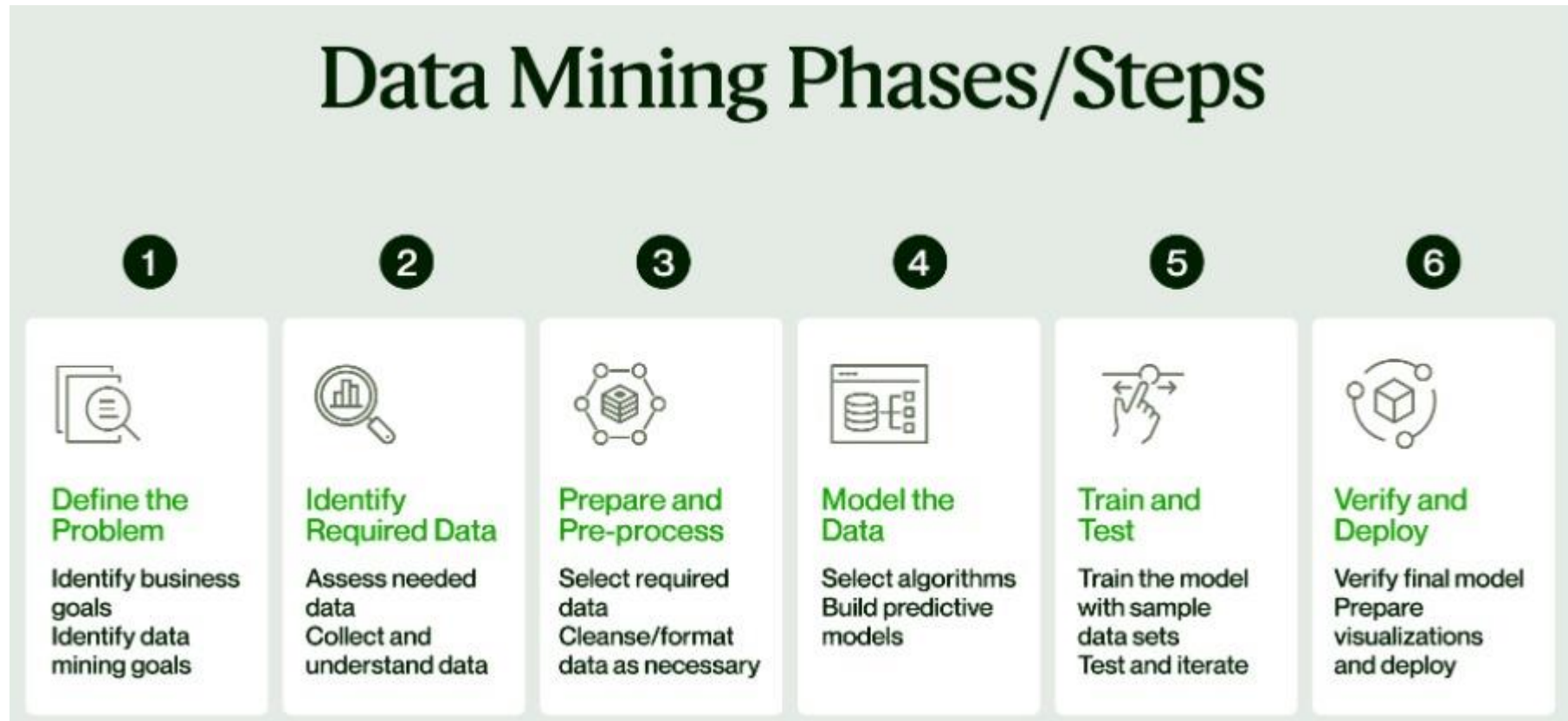
Veracity related to accuracy or trustworthy

Big data system as Hadoop distributed file systems and Spark

Data mining

- ❑ Data mining is the process of sorting through large data sets to identify patterns and extract knowledge from data.
- ❑ Data mining techniques and tools enable enterprises to predict future trends and make more-informed business decisions.

Data mining process



Data gathering

- ❑ Relevant data for an analytics application is identified and assembled.
- ❑ The data may be located in different source systems, that contain a mix of structured and unstructured data.
- ❑ External data sources may also be used. Wherever the data comes from, a data scientist often moves it to a data lake for the remaining steps in the process.

Data preparation.

- ❑ This stage includes a set of steps to get the data ready to be mined.
- ❑ It starts with data exploration, profiling and pre-processing, followed by data cleansing work to fix errors and other data quality issues.
- ❑ Data transformation is also done to make data sets consistent.

Mining the data

- ❑ Once the data is prepared, a data scientist chooses the appropriate data mining technique and then implements one or more algorithms to do the mining.
- ❑ In machine learning applications, the algorithms typically must be trained on sample data sets to look for the information being sought before they're run against the full set of data.

Data analysis and interpretation

- ❑ The data mining results are used to create analytical models that can help drive decision-making and other business actions.
- ❑ The data scientist or another member of a data science team also must communicate the findings to business executives and users, often through data visualization and the use of data storytelling techniques.

Types of data mining techniques

Association Rule

- Apriori
- Eclat

Regression

- Simple Linear Regression
- Multiple Linear Regression
- Support Vector Regression
- Decision Tree Regression
- Random Forest Regression

Classification

- KNN
- Support Vector Machine
- Kernel SVM
- Naïve Bias
- Decision Tree Classification
- Random Forest Classification

Clustering

- K-means Clustering
- Hierarchical Clustering

Types of data mining techniques

Association rule mining: identify relationships between data elements.

Classification: assigns the elements in data sets to different categories. It determines a predefined number of classes because data is labeled.

Clustering: data elements that share particular characteristics are grouped together into clusters based on similarity.

Regression: calculating predicted values based on a set of variables.

Thank you!

